

(b)

PV panel charges a 24 V battery: 24 V for the PLC and the belt motor; step-down converters 12 V for the stepper driver and 5 V for logic

machine layer

intake sensor X0
IR jam beam X2
E-stop (NC) X3

from Pi relays:
pause X4
bins full X5–X7

PLC
Delta DVP-14SS2 + DVP-08SP
run latch, jam reverse,
PWM speed, bin logic

BTS7960
Y0 PWM / Y1 rev.

24 V worm-gear
motor (belt)

beacon Y4
run lamp Y5

relays K1–K4
(isolated contacts)

vision and sorting layer

Logitech C270
USB camera

USB

phone / laptop
(Wi-Fi)

HTTP

Raspberry Pi 4
YOLOv8n (COCO), 320 px, conf 0.55
tracker, scheduler, bin counters
bin-level ultrasonics, web dashboard

STEP 27
DIR 17

A4988
full step

NEMA 17
+ printed gate
(3 routes)